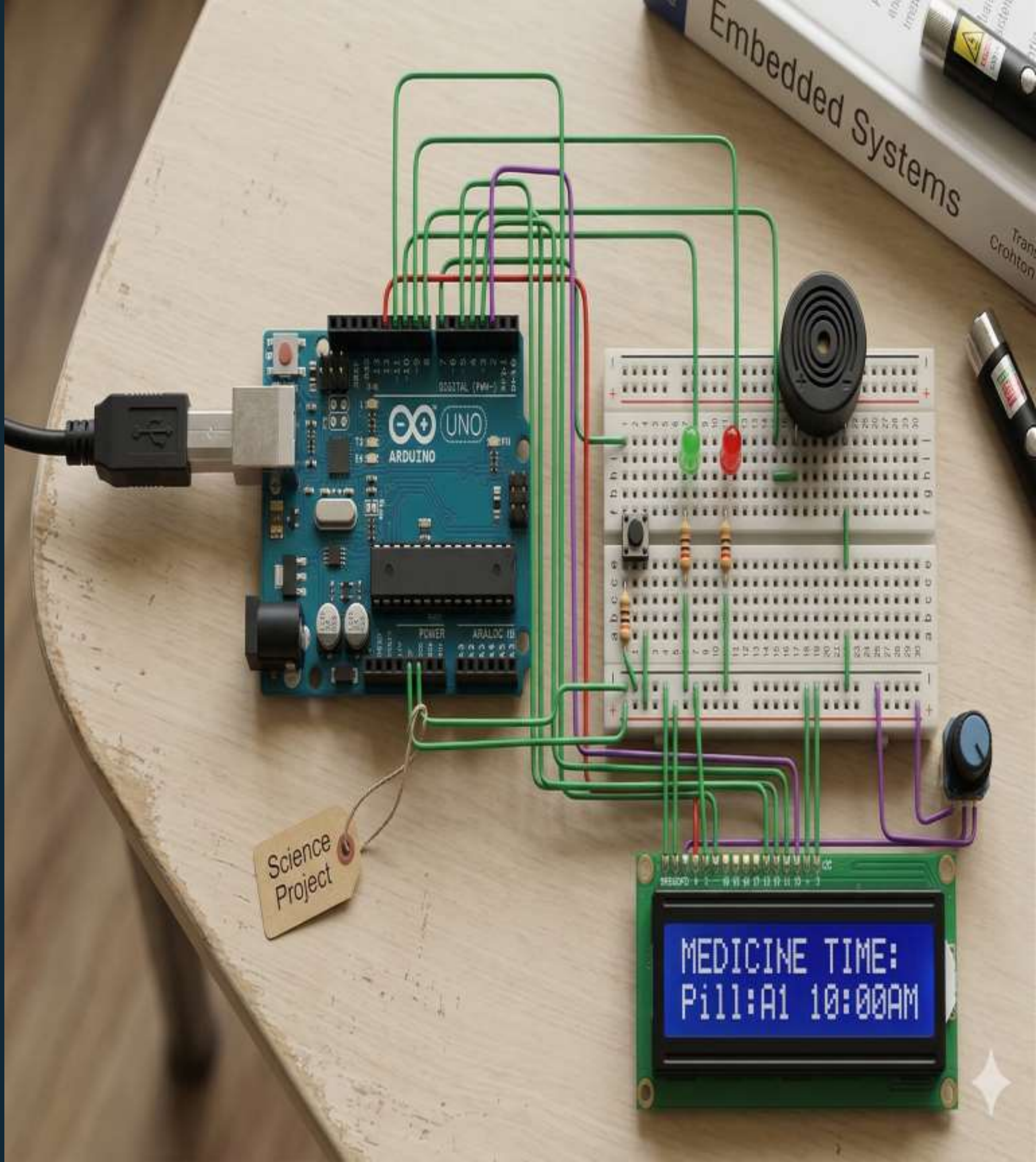
SMART MEDICINE REMINDER SYSTEM

Arduino-Based Design and Simulation Using TinkerCAD

BIOMEDICAL ENGINEERING 3

MICRPROCESSORS

Project Team: SHECAN



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INTRODUCTION

What This Project Does

- This project focuses on designing a smart medical reminder system for elderly people
- This Arduino-based system automatically reminds users to take medication at preset times.
- It aims to improve medication adherence and health outcomes

Problem Statement

Medication Non-Adherence

Approximately 50% of patients fail to take medications as prescribed, leading to worsening health conditions and increased healthcare costs.

Cognitive Challenges

Memory issues, particularly among elderly patients and those with chronic conditions, make it difficult to maintain consistent medication schedules.

Lack of Affordable Solutions

Existing smart pill dispensers can cost hundreds of dollars, putting them out of reach for many patients who need reminder systems.

Background

Why This Matters

125K

Annual Deaths

Caused by medication non-adherence in the United States

$300B

Healthcare Costs

Attributed to medication non-compliance annually

70%

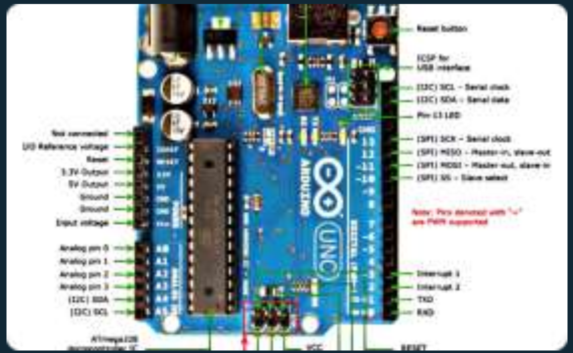
Improvement Potential

Patients who use reminder systems show better adherence

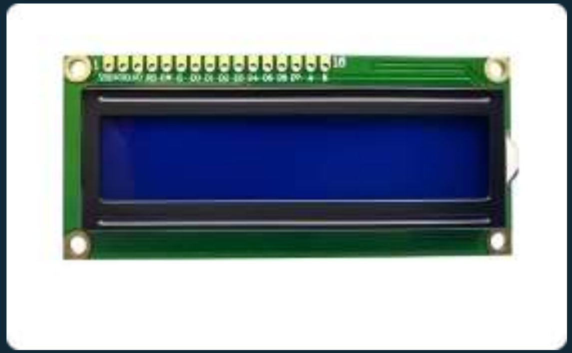
Key Concepts

Arduino Microcontroller

An open-source electronics platform based on easy-to-use hardware and software. It reads inputs (buttons, sensors) and controls outputs (LEDs, displays, buzzers) through programmed instructions.



Arduino Uno



LCD Display

Push Buttons

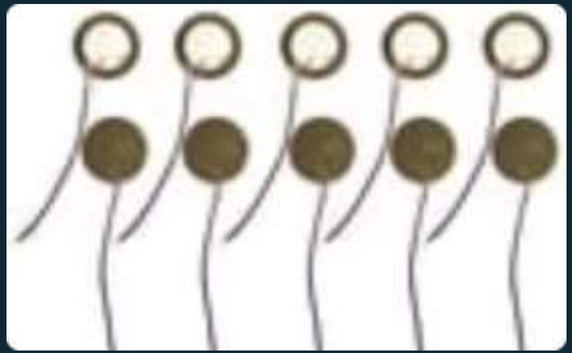
LED Indicator

Digital I/O Control

Arduino pins can be configured as inputs to read button states or outputs to control LEDs and buzzers. Each pin operates at 5V logic levels.

Timing Functions

The millis() function tracks elapsed time without blocking program execution, enabling non-blocking delay-based scheduling for medication reminders.



Buzzer

Project Objectives

01

Program Arduino Microcontroller

Develop embedded C code to manage scheduled medication reminders with configurable timing parameters

02

Interface Push Buttons

Implement button inputs for users to set and adjust reminder times through intuitive interface

03

Display Information on LCD

Configure 16x2 character LCD to show medication schedules, current time, and reminder messages

04

Generate Alert Signals

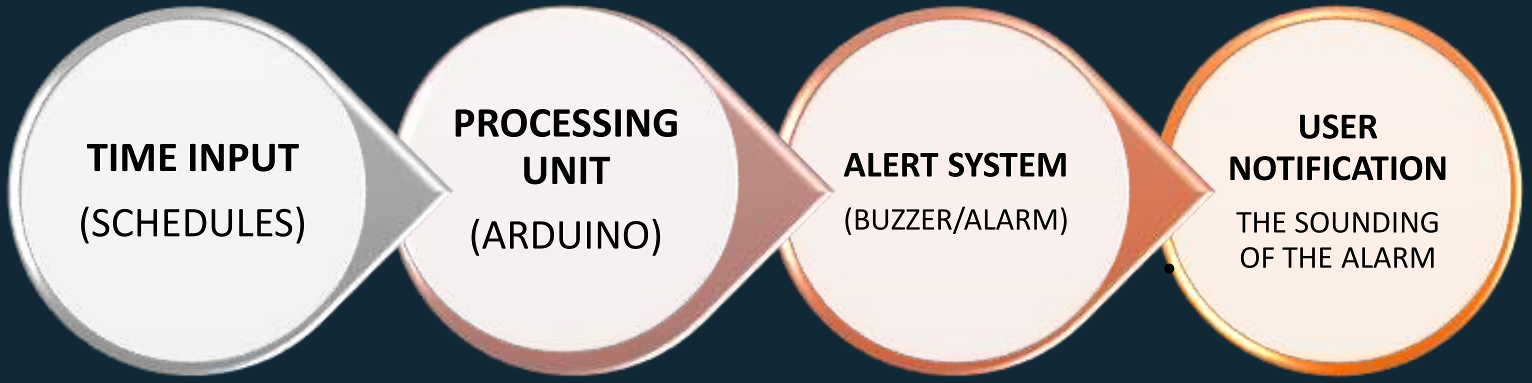
Activate buzzer for audible alerts and LED indicators for visual notifications when medication time arrives

05

Simulate System in TinkerCAD

Test complete system functionality using Arduino simulation platform before hardware implementation

Block Diagram



System Architecture

Arduino Uno

Main controller executing programmed logic

Push Buttons

User input for time configuration(Push buttons for time setting and system control)

LCD Display

Visual feedback of schedules(16x2 LCD shows medication schedule and status messages)

Buzzer & LED

Audio and visual alerts(Buzzer and LED provide audible and visual notifications)

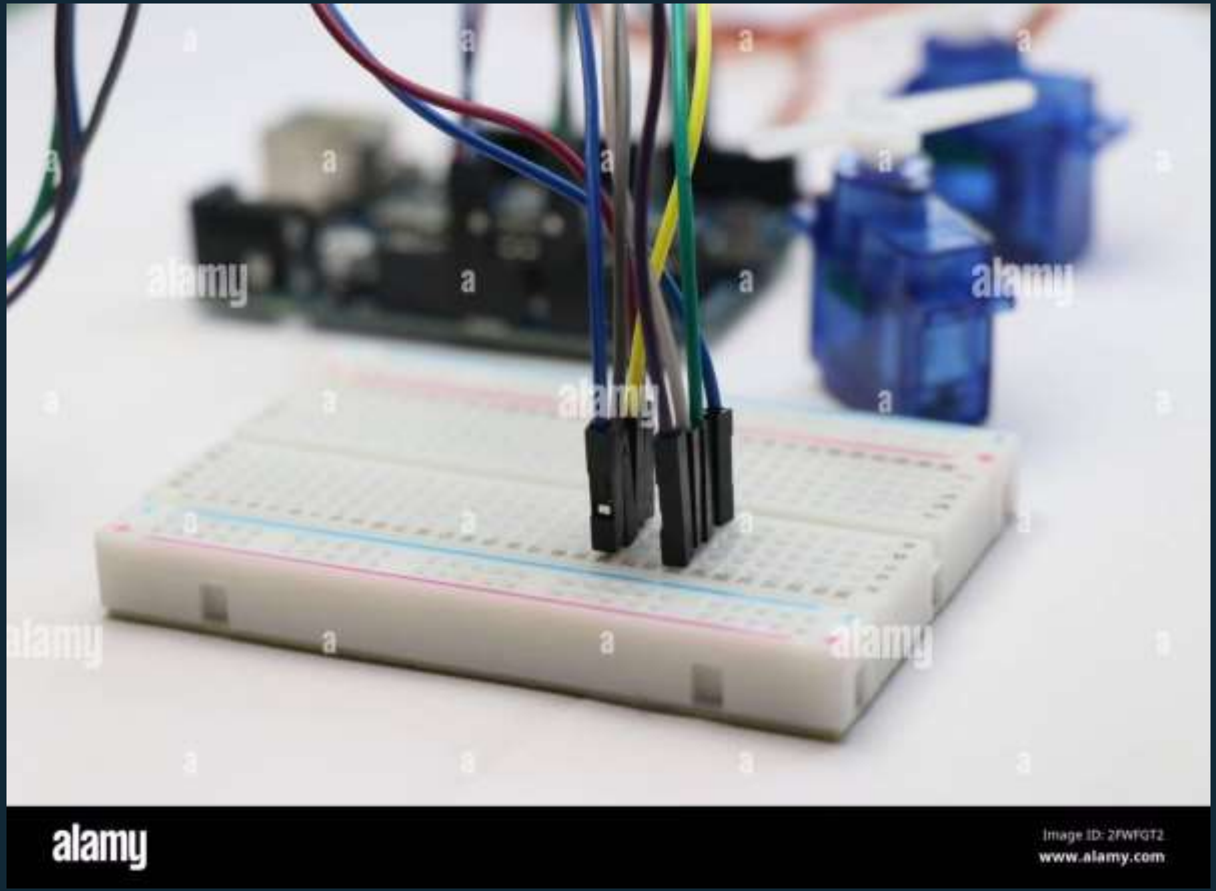
IMPLEMENTATION DETAILS

Hardware Components

- Arduino Uno microcontroller
- 16x2 LCD display (1602)
- Push buttons (4 units)
- Buzzer (active or passive)
- LED indicators
- Resistors (220Ω, 10kΩ)
- Breadboard and jumper wires

Software Tools

- TinkerCAD Circuits (simulation)
- Arduino IDE (programming)
- Embedded C language
- Arduino LiquidCrystal library



Key Code Functions

millis() Timing
Non-blocking delay implementation

digitalRead()
Button state detection

LiquidCrystal
LCD screen control

tone() Function
Buzzer sound generation

```cpp
#include <LiquidCrystal.h>

// ── Pin Definitions ──────────────────────────────
const int BUZZER_PIN    = 8;
const int BUTTON_PIN    = 7;
const int GREEN_LED_PIN = 10;  // Lights up when dose is taken
const int RED_LED_PIN   = 9;   // Lights up when dose is missed

// LCD: RS, EN, D4, D5, D6, D7
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);

// ── Pill Schedule ──────────────────────────────
// Simulated times in seconds from program start (for TinkerCAD
// Alarm 1 = 5s,  Alarm 2 = 15s,  Alarm 3 = 25s
const int NUM_ALARMS = 3;
const long ALARM_TIMES[NUM_ALARMS]   = {5, 15, 25}; // seconds f
const char* ALARM_LABELS[NUM_ALARMS] = {"Morning Pill", "Noon Pi

// ── State Variables ──────────────────────────────
int  currentAlarm   = 0;
bool alarmActive    = false;
int  missedDoses    = 0;
long alarmStartTime = 0;
const long RESPONSE_WINDOW = 10000; // 10 seconds to respond (us

```

```cpp
// ── Setup ──────────────────────────────
void setup() {
  pinMode(BUZZER_PIN,    OUTPUT);
  pinMode(BUTTON_PIN,    INPUT_PULLUP); // LOW = pressed
  pinMode(GREEN_LED_PIN, OUTPUT);
  pinMode(RED_LED_PIN,   OUTPUT);

  digitalWrite(GREEN_LED_PIN, LOW);
  digitalWrite(RED_LED_PIN,   LOW);

  lcd.begin(16, 2);
  lcd.print("Pill Reminder");
  lcd.setCursor(0, 1);
  lcd.print("  Starting up...");
  delay(2000);
  lcd.clear();
}

// ── Main Loop ──────────────────────────────
void loop() {
  long now = millis() / 1000; // current seconds since start

  // 1. Trigger alarm when scheduled time is reached
  if (currentAlarm < NUM_ALARMS && now >= ALARM_TIMES[currentAla
    triggerAlarm(currentAlarm);
  }

  // 2. While alarm is active, wait for button press or timeout
```



```cpp
  // 2. While alarm is active, wait for button press or
```



```cpp
  if (alarmActive) {
    buzz();

    // Button pressed → dose taken
    if (digitalRead(BUTTON_PIN) == LOW) {
      doseTaken();
    }

    // Time window expired → missed dose
    if (millis() - alarmStartTime > RESPONSE_WINDOW) {
      doseMissed();
    }
  }

  // 3. Idle screen when no alarm
  if (!alarmActive && currentAlarm < NUM_ALARMS) {
    showIdleScreen(currentAlarm);
  }

  // 4. All doses done
  if (currentAlarm >= NUM_ALARMS && !alarmActive) {
    showDoneScreen();
  }

```

```cpp
  delay(100);
}

// ── Functions ──────────────────────────────
void triggerAlarm(int index) {
  alarmActive    = true;
  alarmStartTime = millis();

  digitalWrite(GREEN_LED_PIN, LOW);
  digitalWrite(RED_LED_PIN,   LOW);

  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("!! TAKE PILL !!");
  lcd.setCursor(0, 1);
  lcd.print(ALARM_LABELS[index]);
}

void buzz() {
  tone(BUZZER_PIN, 1000); // Change 1000 to your desire
  delay(300);
  noTone(BUZZER_PIN);
  delay(200);
}

```



```cpp
void doseTaken() {
  alarmActive = false;
  noTone(BUZZER_PIN);
```

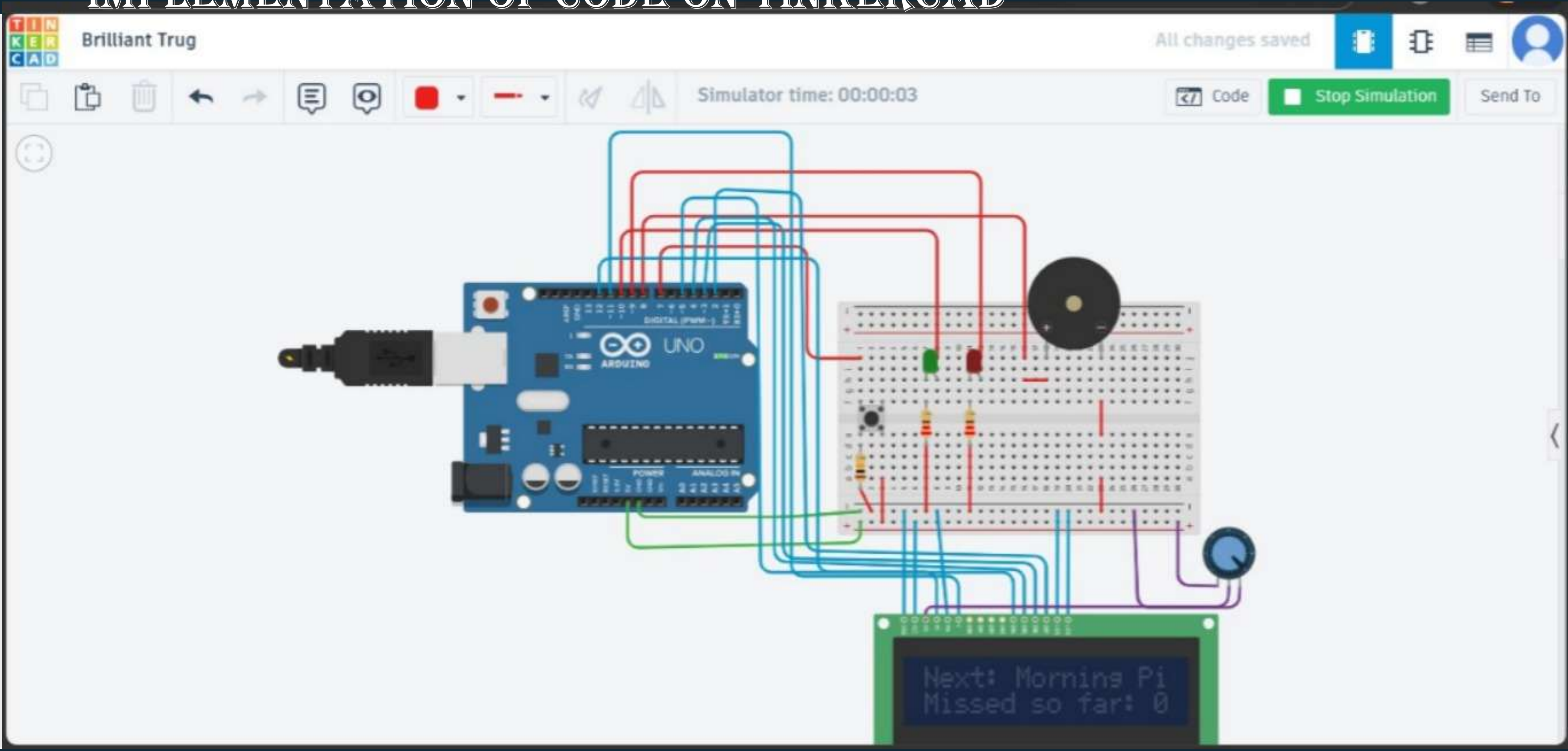


```cpp
void doseTaken() {
  alarmActive = false;
  noTone(BUZZER_PIN);
  digitalWrite(BUZZER_PIN, LOW);

  // Green LED: flash 3 times then stay on bri
  for (int i = 0; i < 3; i++) {
    digitalWrite(GREEN_LED_PIN, HIGH);
    delay(200);
    digitalWrite(GREEN_LED_PIN, LOW);
    delay(150);
  }
  digitalWrite(GREEN_LED_PIN, HIGH); // stay o

  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("  Dose Taken!");
  lcd.setCursor(0, 1);
  lcd.print("Good job :)      ");
  delay(2500);

  digitalWrite(GREEN_LED_PIN, LOW);
  lcd.clear();
  currentAlarm++;
}

```

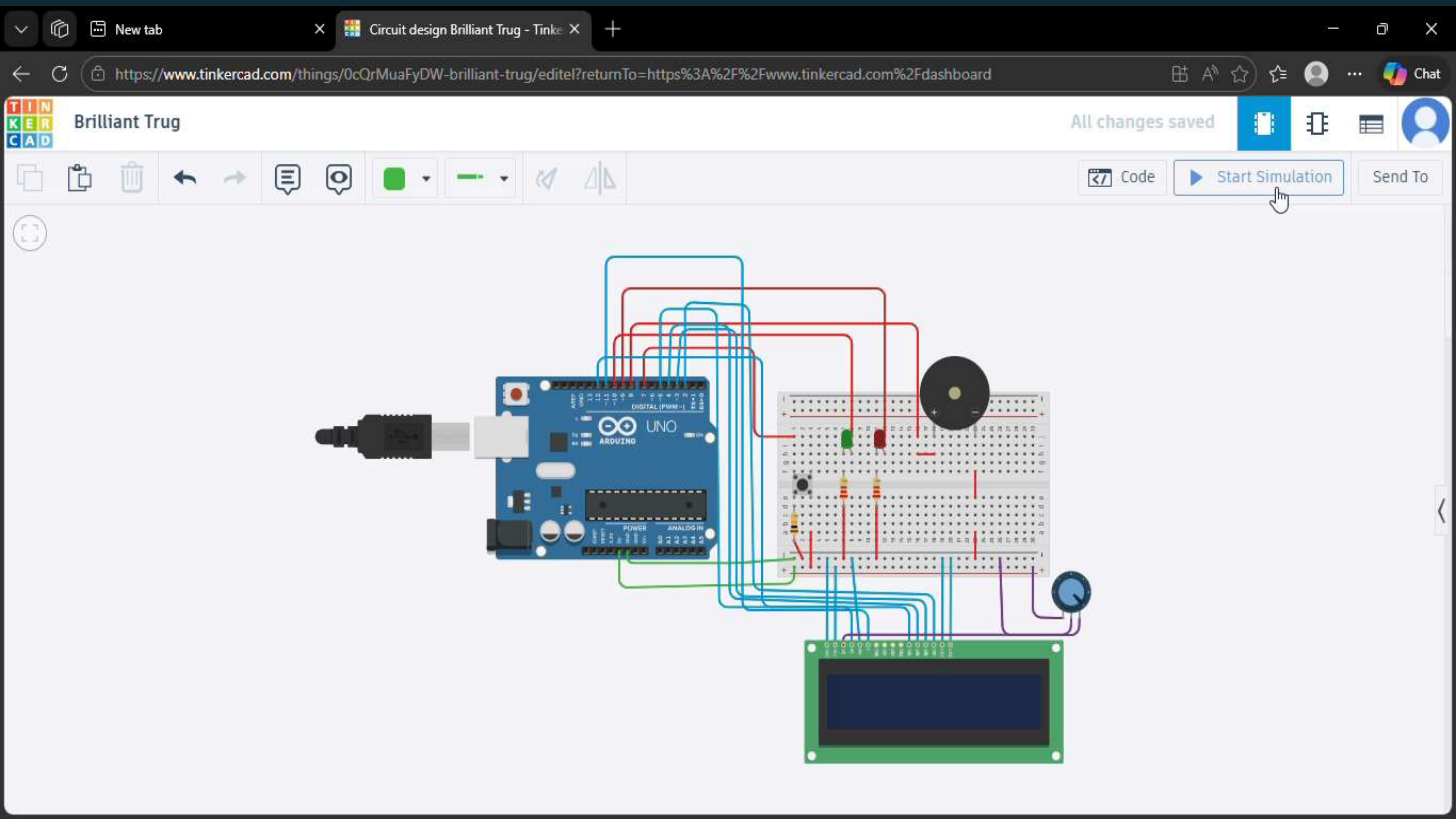
```cpp
void doseMissed() {
  alarmActive = false;
  noTone(BUZZER_PIN);
  digitalWrite(BUZZER_PIN, LOW);
  missedDoses++;

  // Red LED: flash 3 times then stay on brief
  for (int i = 0; i < 3; i++) {
    digitalWrite(RED_LED_PIN, HIGH);
    delay(200);
    digitalWrite(RED_LED_PIN, LOW);
    delay(150);
  }
  digitalWrite(RED_LED_PIN, HIGH); // stay on

  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Dose MISSED!");
  lcd.setCursor(0, 1);
  lcd.print("Missed: ");
  lcd.print(missedDoses);
  delay(3000);

  digitalWrite(RED_LED_PIN, LOW);
  lcd.clear();
  currentAlarm++;
}
```

IMPLEMENTATION OF CODE ON TINKERCAD

```
  lcd.print("Dose MISSED!");
  lcd.setCursor(0, 1);
  lcd.print("Missed: ");
  lcd.print(missedDoses);
  delay(3000);

  digitalWrite(RED_LED_PIN, LOW);
  lcd.clear();
  currentAlarm++;
}

void showIdleScreen(int index) {
  lcd.setCursor(0, 0);
  lcd.print("Next: ");
  lcd.print(ALARM_LABELS[index]);
  lcd.setCursor(0, 1);
  lcd.print("Missed so far: ");
  lcd.print(missedDoses);
}

void showDoneScreen() {
  lcd.setCursor(0, 0);
  lcd.print(" All doses done ");
  lcd.setCursor(0, 1);
  lcd.print("Missed: ");
  lcd.print(missedDoses);
  lcd.print("      ");
}
```



DEMONSTRATION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Chat
TIN KER CAD
Brilliant Trug
All changes saved
Code
Start Simulation
Send To
DIGITAL (PWM~)
UNO
ARDUINO
POWER
ANALOG IN

Results, Challenges & Conclusion

Simulation Success

System successfully simulated in TinkerCAD, demonstrating accurate timing, proper button response, LCD display functionality, and alert generation at preset intervals.

Challenges Encountered

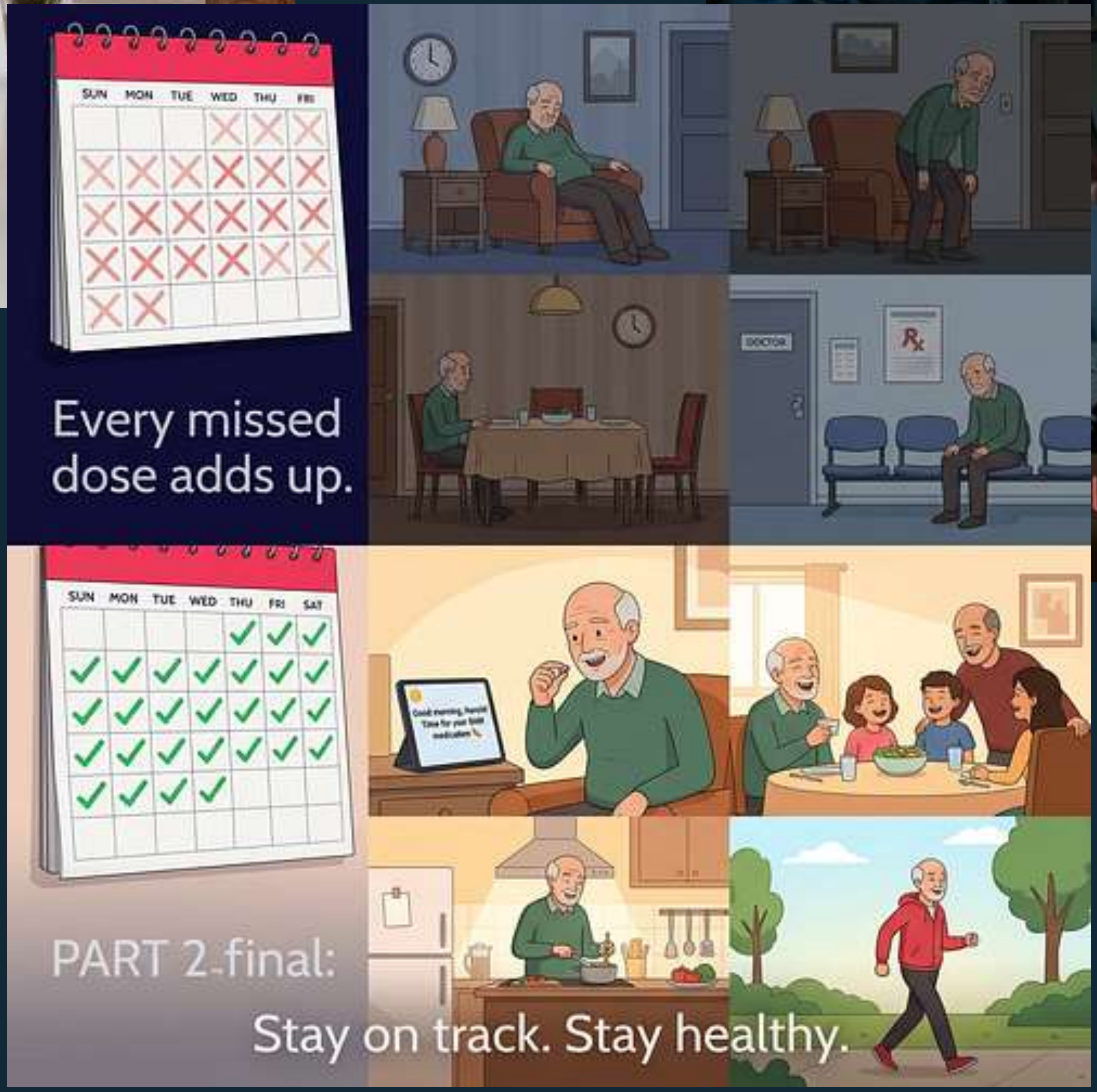
Debouncing button inputs required software delays, LCD initialization timing needed optimization, and buzzer duration had to be carefully calibrated to avoid excessive noise.

Future Improvements

Add EEPROM storage for schedule persistence, implement real-time clock module for accurate timekeeping, integrate Wi-Fi for remote monitoring, and expand to multiple medication schedules.

Project Conclusion

This project successfully demonstrates embedded systems design principles through a practical healthcare application. The system validates microprocessor concepts including digital I/O control, timing functions, and hardware-software integration. The simulation provides foundation for future hardware implementation and real-world testing with end users.



THANK YOU

DON"T FORGET TO SET A REMINDER TODAYYYY..